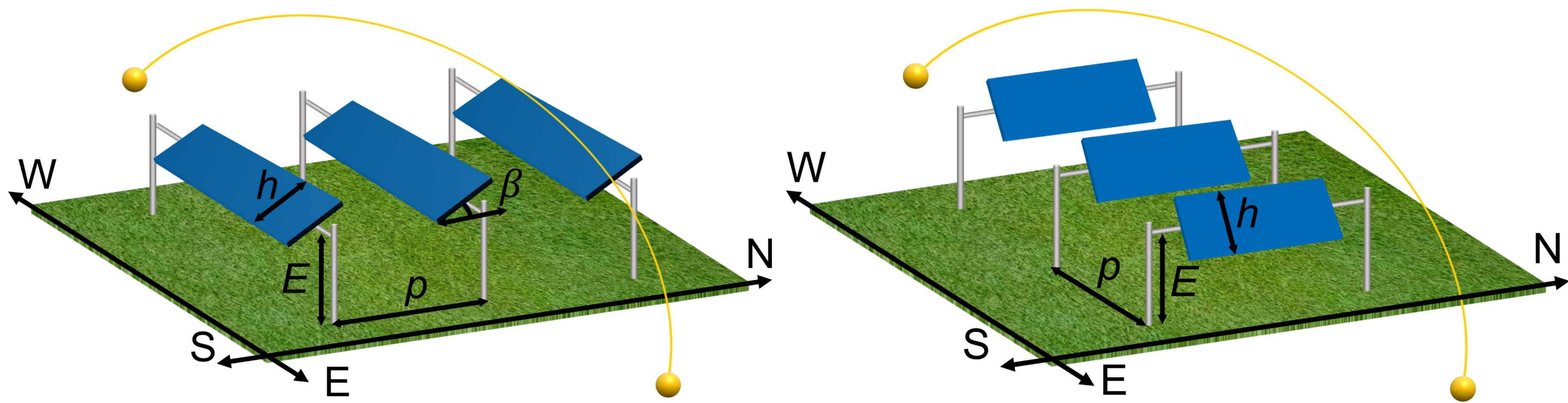


Optimization of Single-Axis Tracking of Photovoltaic Modules for Agrivoltaic Systems

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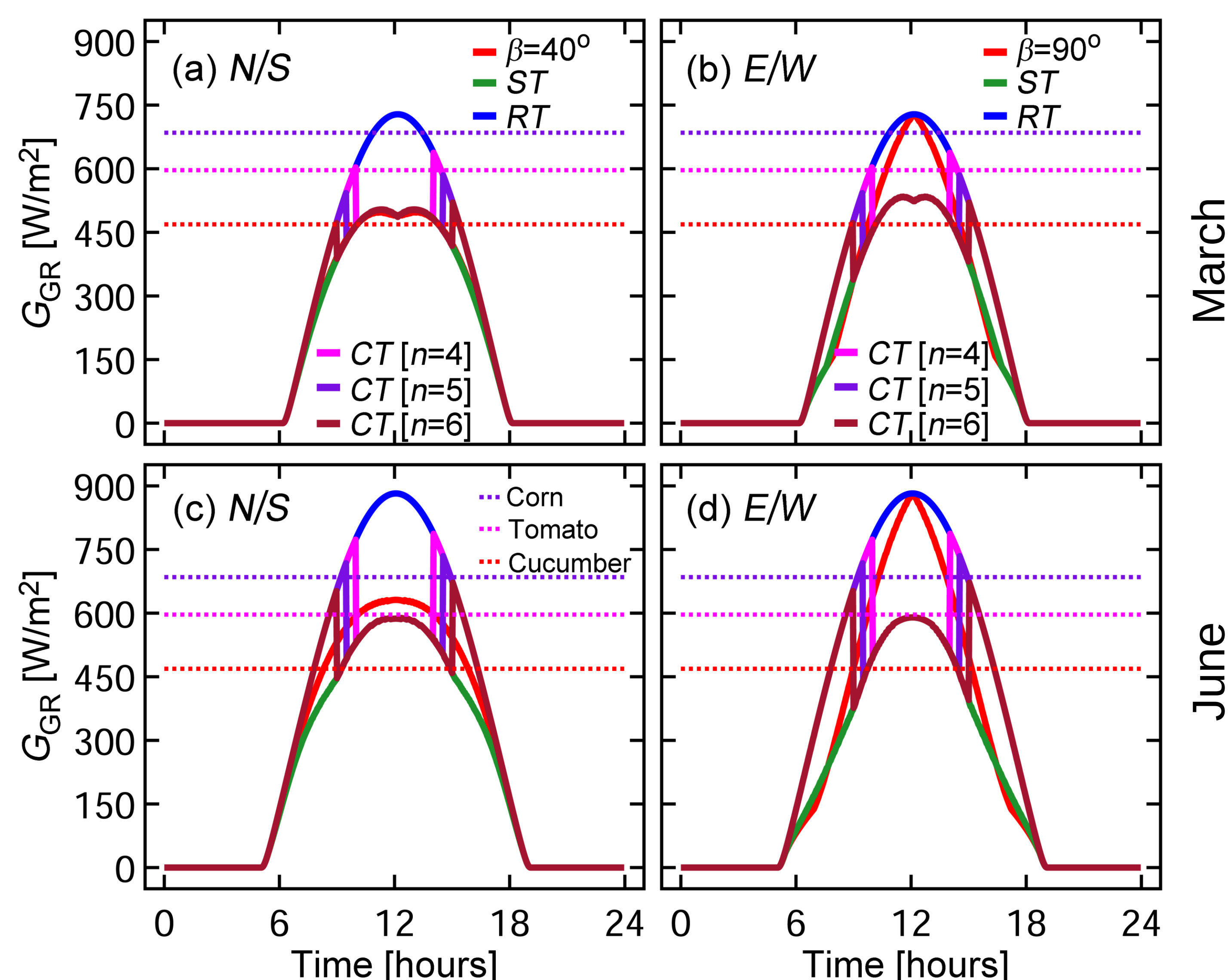
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Introduction



- Agrivoltaic systems (AV) are innovative production systems for dual production of solar energy and agriculture from the same land.
- Tracking schemes that are well-established for standard PV systems may not work well for AV. Need unique customizations for food-energy nexus.
- Standard East/West single axis tracking schemes are explored for bifacial AV farms & compared to non-standard North/South tracking.
- For modeling details, please see [1].

Fixed vs. Single Axis Tracking [Full Array Density]

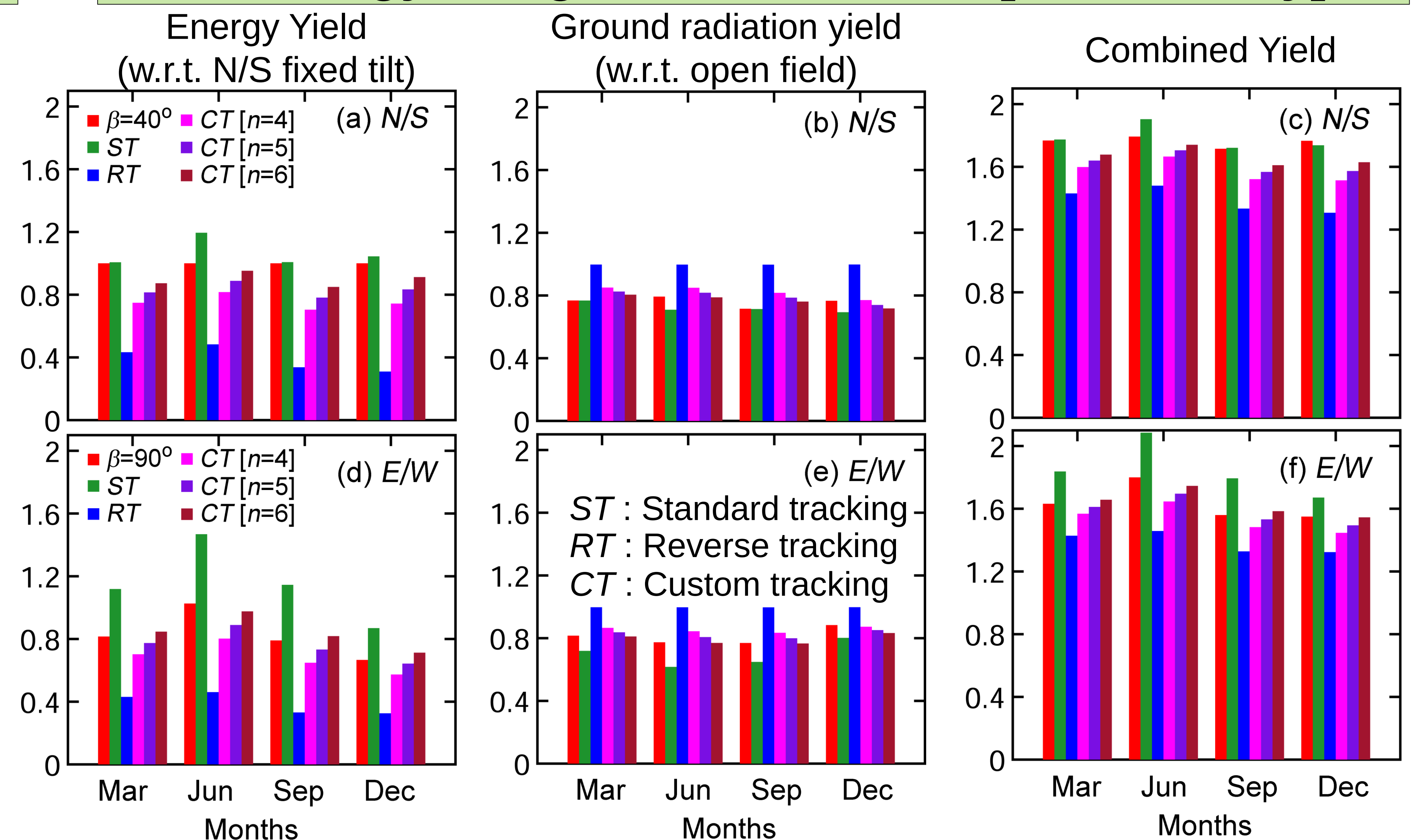


- Each crop has a unique threshold for its photosynthetically active radiation (PAR) needs (shown by dotted lines).
- With RT, transmitted PAR can be over-delivered beyond the crop need.
- Customized (CT) schemes can provide the precise crop needs.

Conclusions

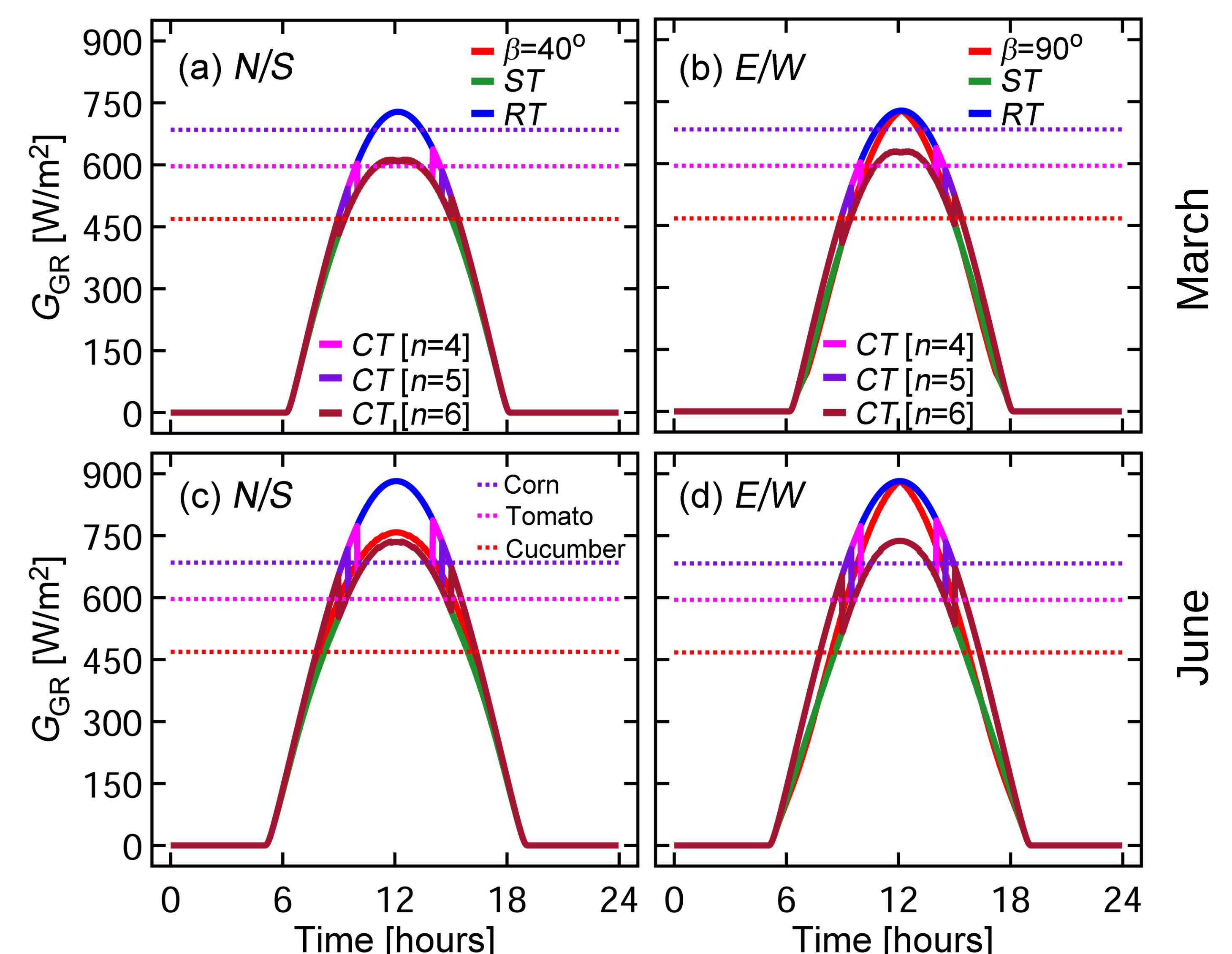
- We propose customized solar tracking schemes that vary between standard sun tracking and reverse sun tracking during the day.
- Crop specific optimizations for the tracking schemes and optimally selected array density can together provide the best food-energy tradeoff.

PV Energy & Light Transmission [Full density]



- Irradiation transmitted to ground is < 80% for fixed tilt (Full density arrays).
- Customized tracking can offer a desired trade-off between PV & crop needs.

Fixed vs. Single Axis Tracking [Half Array Density]



- Reduced PV array density is commonly used for shade sensitive farms
- Optimization of array density & tracking schemes needs to go hand in hand for AV systems with tracking.

References

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- [3] C. Dupraz, et al., *Renewable energy*, vol. 36, no. 10, pp. 2725–2732, 2011.
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